

Evaporation



Pre-Lab Plan Unit Operations Lab # 3

September 30th, 2025

Group 3, Tuesday, PM section

Basil Hashim

Lindsey Meisch

Kendall Johnson

1. Experimental Objective (5 pts)

The goal of this experiment is to optimize a single-effect evaporator containing sugar and water by changing the vacuum pressure. The efficiency of the system is parametrized by capacity and economy, two quantities that can be calculated using the amount of steam and product moving through the system. 

2. Experimental

In this lab, the single-effect evaporator contains several parts described in **Figure 1**. The 10 wt% sugar solution is loaded into a reservoir (labelled #2) and heated using steam to begin the evaporation. Applying a vacuum using the pump (labelled #8) lowers the boiling point of the solution and increases the difference in temperature between the steam and boiling liquid, increasing the rate of heat transfer in the system, affecting the key performance indicators (capacity and economy) that are being considered in this lab. Within the single-effect system, the vapor removed from the boiling mixture is condensed and discarded into another flask (labelled #6). Collecting both this and the steam used to heat the system allow for the mass flow through the system to be quantified, which is essential for further calculations.

2.1 Apparatus (5 pts)

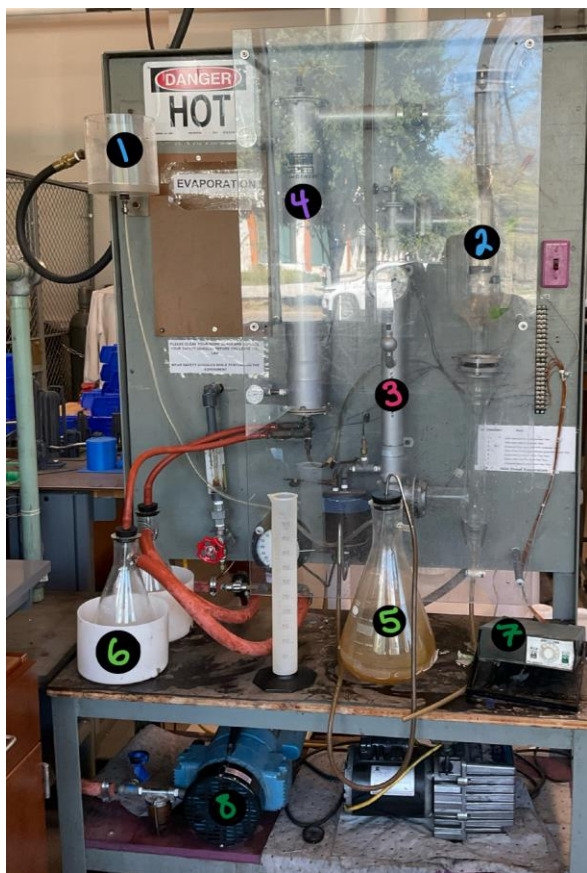


Figure 1: Apparatus with color coded labels.

Number/Color	Apparatus Part
1 – Blue	Make-up DI Water Reservoir
2 – Blue	Sugar Water Reservoir
3 – Pink	Evaporator
4 – Purple	Condenser
5 – Light Green	Flask for Steam Condensate
6 – Light Green	Flask for Water Vapor Product
7 – Dark Green	Thermocouple Reader
8 – Dark Green	Vacuum Pump



2.2 Materials and Supplies (5 pts)



Material/Supplies	Purpose/Specifications
Sugar	To be used for the 10wt% sugar/DI water mixture

DI Water	Used for both the 10wt% sugar/DI water mixture and the fresh water inlet in the make-up reservoir
Thermocouples	Used to read different temperatures across the apparatus
Steam	Used in the evaporator for heat exchange
Cooling Water	Used in the condenser for heat exchange

2.3 Experimental Plan (10 pts)



- First, drain and inspect the separator and verify that all valves are operable and product receivers are clean.
- Create the feed by mixing a 10 wt% sucrose solution with 333g of sugar mixed in 3L of water.
- Use a funnel to fill the separator to the marked level with the prepared solution.
- Fill the makeup tank with deionized water.
- Open the valve to run the cooling water, then the valve to introduce the steam, then start the vacuum pump.
- Once the solution has circulated and the temperatures have stabilized, start measurements.
- Empty the steam-condensate flask, ensuring a product-water flask is empty.
- Collect steam condensate and distillate simultaneously for 5-10 minutes.
- Record the volume and time to get the flow rates.
- Wait 5-10 minutes and repeat.
- Record cooling-water flow and inlet and outlet temperatures at the condenser for an energy-balance cross check.
- Change the vacuum setting and repeat the above steps at three distinct vacuum settings.
- Compute capacity and economy then graph each vs absolute pressure.
- Relate the temperature change between steam and boiling solution.

The independent variable is the system pressure, which changes the boiling point and the temperature as the driving force. The target setpoints are the three vacuum settings of low, medium and high, within equipment limits and the actual gauge value will be recorded for each run. The controlled conditions are the feed composition at 10 wt% solution, make up DI water which is added to keep the separator level constant so composition stays effectively constant. Steam on the shell side and cooling water flow to the condenser maintained at steady settings are also controlled conditions.

Dependent variables include capacity, economy, boiling temperature and cooling water flow and temperature change. Capacity will be measured from the timed volume collection. Economy will be measured from simultaneous timed volumes of product and steam condensate. The boiling temperature of the solution and steam/condensing side temperature from the

thermocouples and instrumentation will be used to compute the change in temperature. The cooling water flow and change in temperature will be collected at the outlet for a steam duty check.

Expected ranges will be expected for the vacuum at three points spanning a reasonable range determined by the equipment limitations, boiling temperature which decreases with higher vacuum while noting boiling point elevation vs pure water, and the change in temperature (steam boiling) increases as vacuum deepens, driving higher capacity (theory expectation).

Replicates include $n = 2$ at each vacuum setpoint (6 sets of data total). Timing windows will be greater than or equal to 1-2 minutes to reduce relative error. Reported estimates are the mean of replicates per setpoint, a 95% confidence interval on mean capacity and economy using t-distribution. The goodness of fit will be on a linear regression of Capacity vs ΔT , report slope, intercept, R^2 , and inspect residuals for curvature. Capacity vs Absolute Pressure and Economy vs absolute Pressure with trend commentary will also be presented.

Receiver graduations will be verified for the volumes at one point with a graduated cylinder and calibration uncertainty will be recorded. A stopwatch will be cross-checked against a reference clock and a sanity-check will be done with the thermocouples at ambient temperature and offsets will be noted.

The heat transfer model, using $q = UA\Delta T$ as the core relation, apparent UA will be inferred from measured evaporation duty $q \approx \dot{m}_{\text{product}}\lambda$ at each setpoint and Capacity $\propto \Delta T$. The energy balance will be calculated with a steady-state balance around the exchanger/separators using steam condensed and condenser water heat up, $\dot{m}_{\text{cw}}c_p\Delta T$ as a consistency check and any losses will be discussed. Capacity and Economy will be defined and interpreted per manual and observed trends vs vacuum will be compared to expectations.

The measured boiling temperature vs absolute pressure will be compared and boiling point elevation and liquid head effects that shift from pure water predictions will be noted.

Project Safety Evaluation (10 pts)



This experiment involves operating a single-effect evaporator under vacuum conditions to concentrate a sugar solution, which introduces several potential hazards. The primary hazards are with the pressurized steam, hot surfaces, vacuum operation and glassware under reduced pressure. The steam operates below 40 psia with condensing steam at elevated temperatures near 100°C, presenting burn risks when handling valves or condensate lines. The boiling solution, maintained at around 60-90°C, depending on vacuum level, also poses burn risks if splashing occurs. The vacuum pump reduces the boiling temperature of the solution, but creates the risk of implosion if glassware is flawed or over pressurized. The sugar solution itself is not toxic or dangerous, but dust or spills may cause slipping hazards. Water near electrical components poses a hazard if spilled.

These hazards will be controlled through several measures. All hot steam and condensate lines are insulated and operators must avoid direct contact with steam valves during operation. The evaporator is constructed of metal and designed for moderate vacuum, reducing implosion risk. The system is operated behind a protective plexiglass shield and in a well-ventilated area to ensure that any minor steam or vapor leaks disperse safely. Cooling water flow to the condenser will be maintained at all times to condense vapors, preventing uncontrolled steam release.

	Yes/No	
Potential electrical hazards?	Y	If yes, identify conditions: Water by electric comp.
Potential mechanical hazards?	N	If yes, identify conditions:
Potential pressure hazards?	Y	If yes, identify conditions: Vacuum pump → max 44.7 psia
Potential temperature hazards?	Y	If yes, identify conditions: Steam → 133°C
Hazardous/toxic chemicals involved?	N	If yes, hazards involved:
Gloves required?	Y	If yes, specify type: Heat resistant gloves
MSDS reviewed by all team members?	N	List MSDS sheets reviewed:
Process Parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	133	Generated by steam, used in evaporator
Pressure (psia)	44.7	Generated by Vacuum pump
Safety Controls	yes/no	If yes, Provide description
	Y	Plexiglass covering apparatus, gloves used when touching components using steam.

I have read relevant background material for the Unit Operations Laboratory entitled: “Evaporation” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

<u>Handell Johnson</u>	<u>9/30/25</u>
signature of team member	date
<u>[Signature]</u>	<u>9/30/25</u>
signature of team member	date
<u>Baril Hashin</u>	<u>9/30/25</u>
signature of team member	date

will be measured and converted to absolute pressure for analysis with a safe vacuum range. Temperatures will be tracked with thermocouples to ensure the solution boiling does not exceed the limit of 90°C under vacuum, while steam side temperatures will be near atmospheric boiling conditions. Solution concentration is fixed at 10 wt% sucrose, minimizing any risk of high-viscosity fouling or unexpected foaming.